

PACO ALGAR MUÑOZ

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SUMMARY

Machine Learning Engineer and AI Researcher with experience developing end-to-end AI systems from experimentation to deployment. Combines applied research and software engineering to build efficient, reliable solutions for real-world problems.

PROFESSIONAL EXPERIENCE

Graduate AI/ML Research Engineer <i>KDIS Research Group, University of Córdoba</i>	Jan. 2026 – Present <i>Córdoba, Spain</i>
- Designed and evaluated a hybrid educational recommender integrating collaborative, contextual, textual, and sequential signals, achieving competitive performance against state-of-the-art baselines. - Developed machine learning workflows for industrial predictive maintenance, system diagnostics, and computer vision using real-world data. - Built a multimodal drowsiness detection pipeline combining visual signals with physiological, environmental, and vehicle data. - Refactored predictive-maintenance inference pipelines to improve modularity, maintainability, and integration with existing workflows.	
Software Engineer Intern <i>Magtel</i>	Feb. 2025 – May 2025 <i>Córdoba, Spain</i>
- Optimized and refactored a time series forecasting codebase for energy generation, reducing execution time by 20% and simplifying future development. - Benchmarked forecasting models and translated experimental results into model-selection recommendations for energy generation prediction. - Developed a real-time web application that centralized forecast monitoring and enabled operational teams to inspect model outputs.	

TECHNICAL SKILLS

Programming: Python, Rust, C, C++, TypeScript, JavaScript, Java, SQL
Machine Learning: Deep Learning, Recommender Systems, Computer Vision, Time Series Forecasting, Natural Language Processing, Large Language Models (LLMs)
Frameworks: PyTorch, Lightning, PyTorch Geometric, Scikit-learn, TensorFlow, ONNX, NumPy, Pandas
Generative AI: Retrieval-Augmented Generation (RAG), AI Agents, Multi-Agent Systems, LangGraph, Pydantic AI, Model Context Protocol (MCP)
Engineering & MLOps: Git, Docker, FastAPI, Linux, Data Pipelines, Model Serving, MLflow, Weights & Biases
Languages: Spanish (native), English (fluent)

SELECTED PROJECTS

EDuRec <i>Python, PyTorch, PyTorch Geometric</i> github.com/Pacatro/EDuRec Built a modular hybrid recommender combining collaborative, contextual, textual, and sequential representations with contrastive learning.
InferS <i>Rust, ONNX</i> github.com/Pacatro/infers Developed a lightweight inference engine with GPU-accelerated execution for efficient deployment of ONNX neural networks.
GPoeTry <i>Python, PyTorch</i> github.com/Pacatro/gpoetry Implemented and trained a GPT-style language model from scratch for Spanish poetry generation.
MiniNN <i>Rust</i> github.com/Pacatro/mininn Created a minimal deep learning framework with tensors, automatic differentiation, layers, and optimizers.

EDUCATION

M.Sc. in Artificial Intelligence Research <i>Menéndez Pelayo International University</i>	Oct. 2025 – Jul. 2026 <i>Online</i>
B.Sc. in Computer Science <i>University of Córdoba</i>	Sep. 2021 – Jun. 2025 <i>Córdoba, Spain</i>
Erasmus Exchange in Software Development <i>Munster Technological University</i>	Sep. 2024 – Dec. 2024 <i>Cork, Ireland</i>